

Question:

What is the order of the element 5 in  $U_8$ ?

Answer Choices:

- A. 3
- B. 4
- C. 1
- D. 5
- E. 9
- F. 2
- G. 8
- H. 0
- I. 6
- J. 7

Reasoning + Answer:

The question asks for the order of the element 5 in the group  $U_8$ .

First, understand what  $U_8$  is:

$U_n$  denotes the multiplicative group of units modulo  $n$ , i.e., the set of integers between 1 and  $n-1$  that are coprime to  $n$ , under multiplication modulo  $n$ .

So,  $U_8$  is the multiplicative group of units modulo 8.

Step 1: List the elements of  $U_8$

Numbers from 1 to 7 that are coprime to 8:

8 is  $2^3$ , so numbers coprime to 8 are those not divisible by 2.

Thus:  
1, 3, 5, 7

So  $U_8 = \{1, 3, 5, 7\}$

Step 2: Find the order of the element 5 in this group.

The order of an element  $a$  in a group is the smallest positive integer  $k$  such that  $a^k \equiv 1 \pmod{8}$ .

We compute powers of 5 modulo 8:

$5^1 \pmod{8} = 5$   
 $5^2 = 25 \pmod{8} = 25 - 3 \cdot 8 = 25 - 24 = 1$

So  $5^2 \equiv 1 \pmod{8}$

Therefore, the order is 2.

Answer: The order of 5 in  $U_8$  is 2.

<answer> F </answer>